

Introduction

Using the World Health Organization's (WHO) dataset on life expectancy, this research will investigate how a number of variables, including age structure, employment, education, income or GDP, and health-system indicators relate to who lives longer, across countries and populations. Accordingly, the research question to be considered is: What are the main predictors of individual life expectancy?

In this project, a multiple linear regression model would be used because life expectancy is determined by a complex interplay of multiple interrelated predictors rather than a single factor. By fitting a multiple linear model, we can estimate the individual effect of each variable while also investigating how variables interact with each other, thereby identifying which factors contribute most strongly to variations in life expectancy. The model enables prediction of life expectancy for new observations, providing a useful tool for policymakers and researchers. Additionally, linear regression facilitates hypothesis testing and the construction of confidence intervals for each coefficient, allowing for the evaluation of both statistical significance and model reliability.

There is empirical support for life expectancy being multidimensional. Wirayuda and Chan (2021) found that sociodemographic indicators (education level, literacy level, and socioeconomic factors) and macroeconomic indicators (GDP per capita and income inequality) influence life expectancy significantly; health resource variables (public health expenditures and the number of health care workers) were important in the long term of population life expectancy. Similarly, Afshin et al. (2022) examined the role of 51 health risk factors on life expectancy and demonstrated that removing major behavioral and physiological risks, like smoking and high blood pressure, would increase life expectancy by about five to six years. This highlights that lifestyle management and chronic disease management were also important determinants of health. In addition, a path analysis of socioeconomic variables in Bangladesh noted that education, fertility, and socioeconomic variables are some of the strongest predictors of life expectancy (Nandi et al., 2023).

Together, these examples suggest that life expectancy is not the result of any single determinant, but rather the result of a variety of demographic, social, economic, and health system based variables that interact in various ways. Understanding these relationships through a multiple linear regression framework can help identify priority areas for intervention and guide evidence-based policy decisions aimed at improving population health and reducing life expectancy disparities.

Data Description

The dataset was collected by the World Health Organization (WHO) through its Global Health Observatory (GHO). The GHO records health data each year from official country reports and national health ministries to compare indicators such as life expectancy, vaccination rates, and mortality across countries (World Health Organization, 2015). The

dataset was later made public for research on Kaggle and GitHub (JayVal34, n.d.; World Health Organization, 2015). The released version already combined multiple yearly files. Some smaller countries had missing data, mainly for GDP, population, or Hepatitis B coverage, so those cases were removed to keep only complete records (World Health Organization, 2015; JayVal34, n.d.).

The response variable, Life Expectancy (years), measures how long a newborn is expected to live under current conditions. It is continuous and ranges from about 36.3 to 89 years, with an average of about 69 years. Because the dataset includes different countries and years, each observation can be treated as independent, making a multiple linear regression appropriate for analysis.

The *summary_table* presents the key predictor variables. Adult Mortality and GDP show wide ranges, meaning these variables vary greatly and reflect large differences in health and income across countries. GDP and Alcohol appear right skewed, while Schooling, Polio, and Income composition of resources are more balanced.

For the interaction term, BMI:Status was chosen because BMI might have an impact on life expectancy differently between developed and developing countries. In wealthier countries, a higher BMI usually means obesity-related health risks. In lower-income nations, a higher BMI can show better nutrition and food access. Adding this interaction allows the model to reflect these two patterns within one analysis.

Ethics discussion

First, from the perspective of trustworthiness and ethical justification within the ethics module, we consider the WHO life expectancy dataset to be ethically sound and highly credible. The main point is that this data originates from the World Health Organization (WHO) and the United Nations, internationally recognized institutions. Moreover, it is not a privately collected or simulated dataset, but rather systematically collected and reported to the WHO by national statistical institutions, indicating its actual collection. Additionally, the dataset on Kaggle includes complete variable definitions, measurement units, and collection notes, with comprehensive metadata that enhances its interpretability.

Second, this dataset has received a lot of upvotes on Kaggle and has been used by many data scientists and analysts for data analysis. This indicates that it is a popular and widely used dataset, indirectly demonstrating its quality and further confirming its reliability and recognition from a practical perspective. The process of collecting and using the data is ethically acceptable. The dataset consists of national level data that has been collected anonymously and does not contain any personal information. Therefore, using this data does not violate any individual's privacy. Since those are national level data and not information collected directly from individuals, individual informed consent is not required to use the data. However, this data set is owned by its original authors, the WHO, and respective national governments. And also, it has been released as open data on Kaggle for anyone's research and analysis purposes. Our project is using this data set for public health research,

primarily exploring factors influencing human lifespan, with objectives aligned with the public interest. Therefore, our use of the data is ethically justified.

Preliminary Results

A multiple linear regression model was fitted using Life Expectancy as the response and nine predictors: Year, Status, Adult Mortality, Alcohol, BMI, Schooling, GDP, Polio, and Income Composition of Resources, along with an interaction term $\text{BMI} \times \text{Status}$. The model was fitted on 2,938 complete observations.

The regression output shows an $R^2 = 0.7757$, meaning the predictors explain about 78% of the variation in life expectancy. Adult Mortality, Schooling, GDP, Polio, and Income Composition of Resources are significant positive predictors of life expectancy. The interaction BMI:Status is significant, which supports the idea that the effect of BMI differs between developed and developing countries. Year was not significant, showing that after accounting for other variables, time alone did not strongly affect life expectancy.

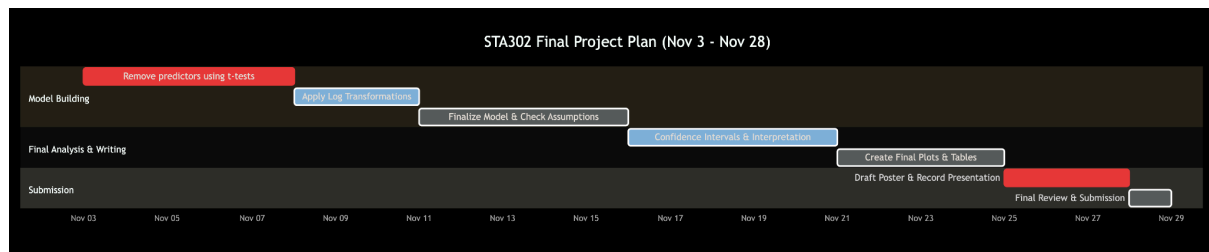
The residual-versus-fitted plot shows no major violations. The points are evenly scattered around zero, which means the assumptions of linearity and constant variance are generally satisfied. The residual-versus-predictor plots also look good overall, though the plots for Adult Mortality and GDP show small signs that could suggest non-linearity or not constant variance. Adult Mortality might have violated linearity assumption and GDP might have violated the constant variance assumption due to a fanning shape present in the plot. The boxplot of residuals by Status shows a much wider spread for developing countries, which could be a big problem. This means the variance might not be constant and there's more variability in those countries. The QQ plot shows a slight deviation in the tails at both ends, implying that the normality assumption may not be fully valid. The Response vs Fitted plot looks fine and suggests a linear relationship. The pairwise plots between predictors show some small patterns, but mostly random scatter, so there might be a bit of overlap between predictors.

Plan

1. Predictor Selection: We will try to remove some of the predictors from the model, for those who has the largest p-value greater than 0.05. After removing every predictor that has the p-value greater than 0.05, we will fit the new model and check the new model's p-values again. We will repeat this process until only significant predictors remain. We may keep a predictor that is not significant if it is very important based on our literature review.

2. Fixing Model Problems: We will check the residual plots after every change to the model. Our preliminary model shows problems with constant variance and Normality. To fix this, we will try a log transformation of the response variable, 'Life.expectancy'. If this does not work well enough, we will also try a log transformation of the GDP predictor. We will continue to check the plots to see if the problems are solved.

3. Final Model and Interpretation: When we have a final model with no clear assumption violations, we will use it to understand the relationships in the population. We will create 95% confidence intervals for the important slopes, such as the one for Status. This will allow us to see the range of possible values for the true relationship in the population.



Bibliography

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